Arquitectura de Robots

Spatial transformations

Dr. Mario Angel Rico Mendez

29 de octubre de 2026

FIME
FACULTAD DE INGENIERÍA MECÁNICA Y ELÉCTRICA



Table of Contents

1 Transformation in relation to the frame

- Transformation in relation to the frame
- Examples

Transformation in relation to the frame

Translation and rotation are transformations performed in relation to a reference system.

If you want to represent the position and orientation of a reference system B that coincides with reference system A and that has been rotated and translated according to it, you must consider whether the rotation was performed first and then the translation, or vice versa.

Section 1.1

Rotation followed by a translation

Rotation about X followed by a translation

1 Transformation in relation to the frame

In the case of performing a rotation on one of the axes of system A followed by a translation, the homogeneous matrices will be as expressed below:

Rotation of an angle theta about X followed by translation of a vector $P(x, y, z)$

$$T((x,\theta),p) = \begin{bmatrix} 1 & 0 & 0 & P_x \\ 0 & c\theta & -s\theta & P_y \\ 0 & s\theta & c\theta & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1}$$

Rotation followed about Y by a translation

1 Transformation in relation to the frame

Rotation of an angle theta about Y followed by translation of a vector $P(x, y, z)$

$$T((y,\theta),p) = \begin{bmatrix} c\theta & 0 & s\theta & P_x \\ 0 & 1 & 0 & P_y \\ -s\theta & 0 & c\theta & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2}$$

Rotation followed about Z by a translation

1 Transformation in relation to the frame

Rotation of an angle theta about Z followed by translation of a vector $P(x, y, z)$

$$T((z,\theta),p) = \begin{bmatrix} c\theta & -s\theta & 0 & P_x \\ s\theta & c\theta & 0 & P_y \\ 0 & 0 & 1 & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

Section 1.2

Translation followed by a Rotation

Translation followed by a Rotation about X

1 Transformation in relation to the frame

When a translation is performed first, followed by a rotation about one of the system's axes, the corresponding homogeneous matrices are presented below.

Translation of a vector $P(x, y, z)$, followed by rotation of an angle theta about X

$$T(p,(x,\theta)) = \begin{bmatrix} 1 & 0 & 0 & P_x \\ 0 & cos\theta & -sin\theta & P_y cos\theta - p_z sin\theta \\ 0 & sin\theta & cos\theta & P_y sin\theta + p_z cos\theta \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4}$$

Translation followed by a Rotation about Y

1 Transformation in relation to the frame

Translation of a vector $P(x, y, z)$, followed by rotation of an angle theta about Y

$$T(p,(y,\theta)) = \begin{bmatrix} cos\theta & 0 & sin\theta & P_x cos\theta + P_z sin\theta \\ 0 & 1 & 0 & P_y \\ -sin\theta & 0 & cos\theta & P_z cos\theta + p_x sin\theta \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5}$$

Translation followed by a Rotation about Z

1 Transformation in relation to the frame

Translation of a vector $P(x, y, z)$, followed by rotation of an angle theta about Z

$$T(p,(z,\theta)) = \begin{bmatrix} cos\theta & -sin\theta & 0 & P_x cos\theta - P_y sin\theta \\ sin\theta & cos\theta & 0 & P_x sin\theta + P_y cos\theta \\ 0 & 0 & 1 & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6}$$



Table of Contents

2 Examples

- Transformation in relation to the frame
- Examples

Section 2.1

Example 1

Example 1

2 Examples

An {B} system has been rotated 90 ° over the O_{XA} axis and subsequently translated a vector ${}^{A}P_{Borg}$(8,-4,12) with respect to the {A} system.

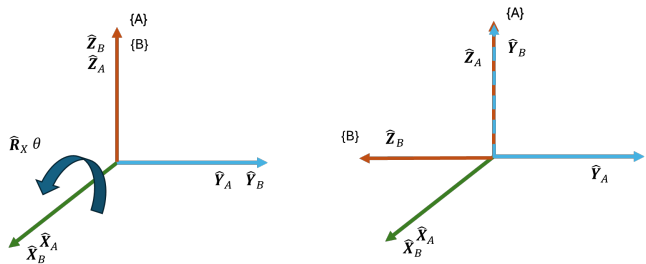
- Calculate the coordinates (r_{XA}, r_{YA}, r_{ZA}) of the vector with the coordinates $r_{XB,YB,ZB}$ (-3,4,-11).

Example solution

2 Examples

An {B} system has been rotated 90 ° over the O_{XA} axis and subsequently translated a vector ${}^{A}P_{Borg}$(8,-4,12) with respect to the {A} system.

Figura 1: *rotation about the X_A axis.*

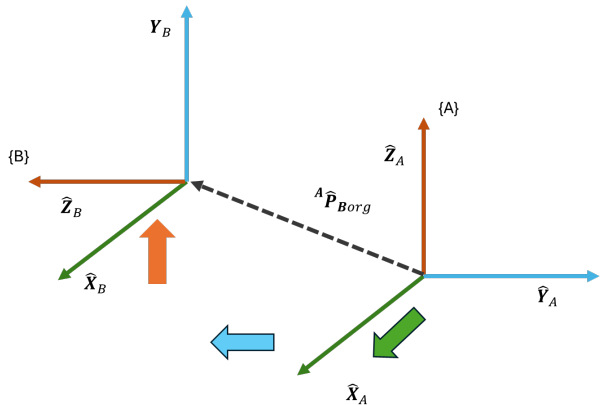


Example solution

2 Examples

An {B} system has been rotated 90 ° over the O_{XA} axis and subsequently translated a vector ${}^{A}P_{Borg}$(8,-4,12) with respect to the {A} system.

Figura 2: *Translation on frame A.*



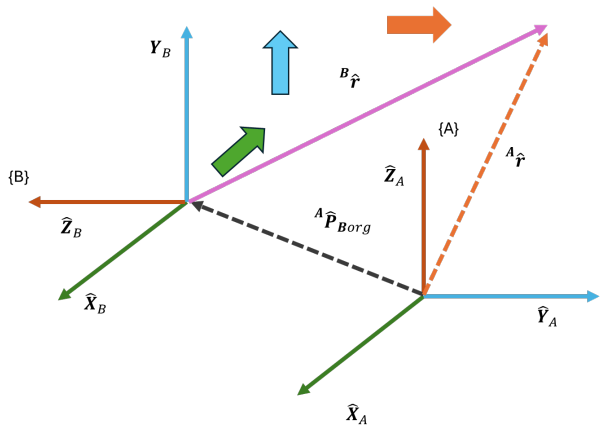
Example solution

2 Examples

An {B} system has been rotated 90 ° over the O_{XA} axis and subsequently translated a vector ${}^{A}P_{Borg}$(8,-4,12) with respect to the {A} system.

- Calculate the coordinates (r_{XA}, r_{YA}, r_{ZA}) of the vector with the coordinates $r_{XB,YB,ZB}$ (-3,4,-11).

Figura 3: *Calculate the coordinates* ${}^{A}r$.



Example solution

An {B} system has been rotated 90 ° over the O_{XA} axis and subsequently translated a vector ${}^{A}P_{Borg}$(8,-4,12) regarding the {A} system.

- Calculate the coordinates (r_{XA}, r_{YA}, r_{ZA}) of the vector with the coordinates $r_{XB,YB,ZB}$ (-3,4,-11).

$$\mathrm{T}\ ((\mathrm{x},\theta),p) = \begin{bmatrix} 1 & 0 & 0 & P_x \\ 0 & c\theta & -s\theta & P_y \\ 0 & s\theta & c\theta & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 8 \\ 0 & c(90) & -s(90) & -4 \\ 0 & s(90) & c(90) & 12 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ 4 \\ -11 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 16 \\ 1 \end{bmatrix}$$

Section 2.2

Example 2

Example 2

2 Examples

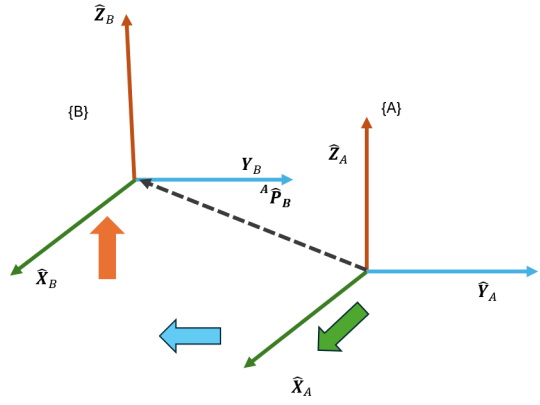
An {B} system has been translated a vector ${}^{A}P_{Borg}$(8,-4,12) with respect to the {A} system, and subsequently rotated 90 ° over the O_{XA} axis.

- Calculate the coordinates (r_{XA}, r_{YA}, r_{ZA}) of the vector with the coordinates $r_{XB,YB,ZB}$ (-3,4,-11).

Example 2 solution

An {B} system has been translated a vector ${}^{A}P_{Borg}$(8,-4,12) regarding the {A} system, and subsequently rotated 90 ° over the O_{XA} axis.

Figura 4: *Translation about the frame A.*

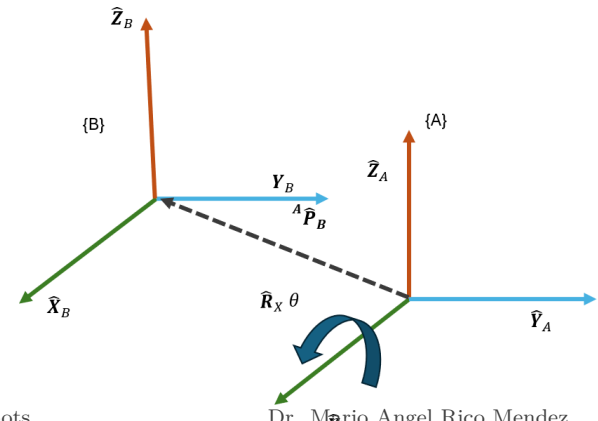


Example 2 solution

2 Examples

An {B} system has been translated a vector ${}^{A}P_{Borg}$(8,-4,12) regarding the {A} system, and subsequently rotated 90 ° over the O_{XA} axis.

Figura 5: *Rotation about the X_A axis.*

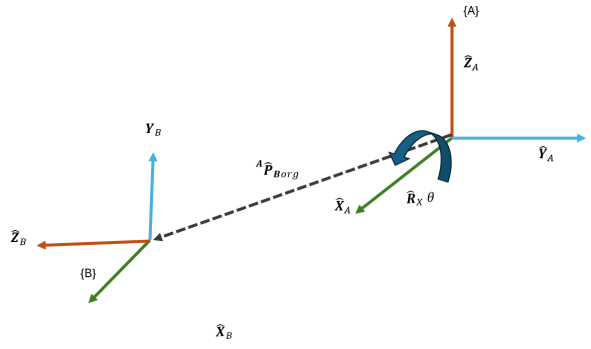


Example 2 solution

2 Examples

An {B} system has been translated a vector ${}^{A}P_{Borg}(8,-4,12)$ regarding the {A} system, and subsequently rotated 90 ° over the O_{XA} axis.

Figura 6: *Rotation about the X_A axis.*



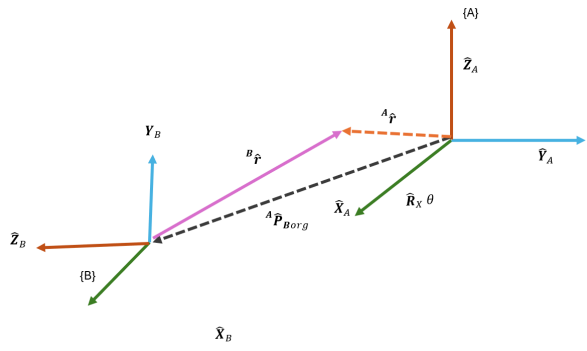
Example 2 solution

2 Examples

An {B} system has been translated a vector ${}^{A}P_{Borg}$(8,-4,12) with respect to the {A} system, and subsequently rotated 90 ° over the O_{XA} axis.

- Calculate the coordinates (r_{XA}, r_{YA}, r_{ZA}) of the vector with the coordinates $r_{XB,YB,ZB}$ (-3,4,-11).

Figura 7: *Calculate the coordinates* ${}^{A}r$.



Example 2 solution

An {B} system has been translated a vector ${}^{A}P_{Borg}$(8,-4,12) with respect to the {A} system, and subsequently rotated 90 ° over the O_{XA} axis.

- Calculate the coordinates (r_{XA}, r_{YA}, r_{ZA}) of the vector with the coordinates $r_{XB,YB,ZB}$ (-3,4,-11).

$$\mathrm{T}\ (\ \mathrm{p},\ (\mathrm{x},\theta)) = \begin{bmatrix} 1 & 0 & 0 & P_x \\ 0 & cos\theta & -sin\theta & P_y cos\theta - p_z sin\theta \\ 0 & sin\theta & cos\theta & P_y sin\theta + p_z cos\theta \\ 0 & 0 & 0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 0 & 0 & 8 \\ 0 & cos(90) & -sin(90) & -4cos(90) - 12sin(90) = -12 \\ 0 & sin(90) & cos(90) & -4sin(90) + 12cos(90) = -4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ 4 \\ -11 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

FIME
FACULTAD DE INGENIERÍA MECÁNICA Y ELÉCTRICA

Q&A

Thank you for your attention

Rodriguez, J. Fundamentos De Robotica Antonio Barrientos.
Craig, J. J. (2006). Introduction to robotics. Pearson Educacion..
Spong, M. W., Hutchinson, S., & Vidyasagar, M. (2020). Robot modeling and control. John Wiley & Sons.